Sparse inversion

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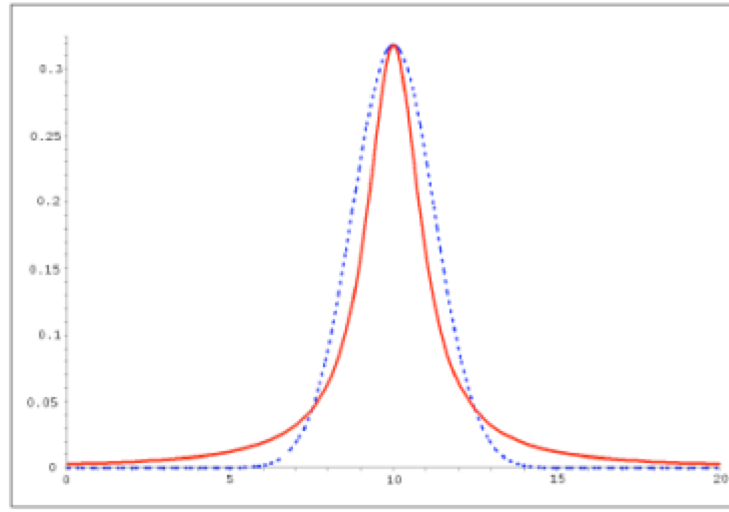
Sparse inversion

Sparseness:

- A few parameters get large values
- Most model parameters stay zero
- Cauchy distribution for the prior

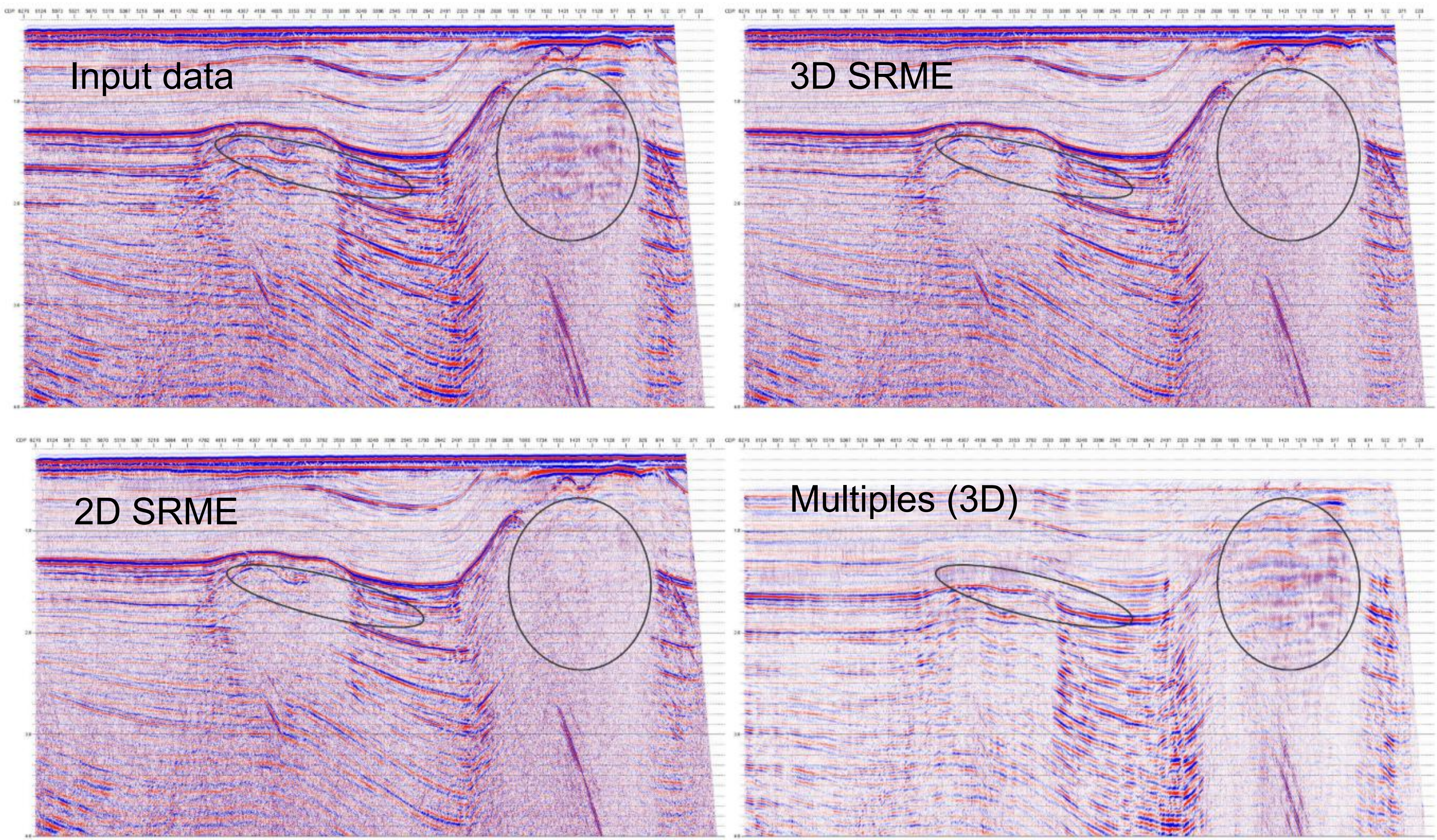
Applications of sparse inversion:

- «Focusing in model space»
- High-resolution Fourier transform
- High-resolution Radon transform
- 3D surface-related multiple elimination (3D SRME)

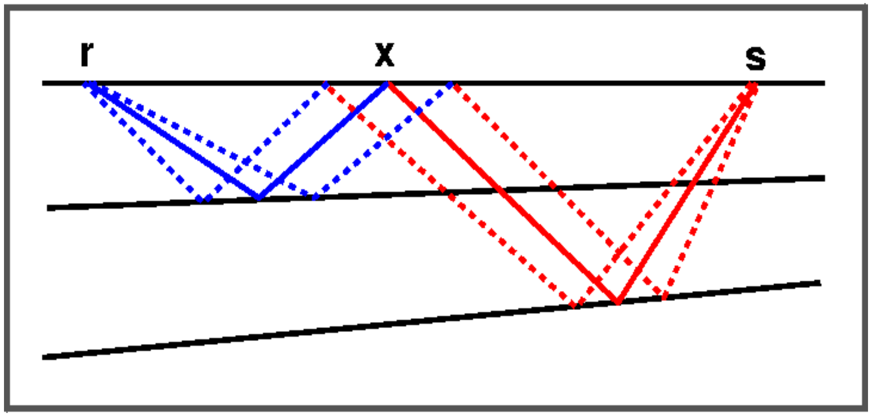


Cauchy distribution (red) and Gaussian distribution (dashed). The Cauchy distribution has infinite variance

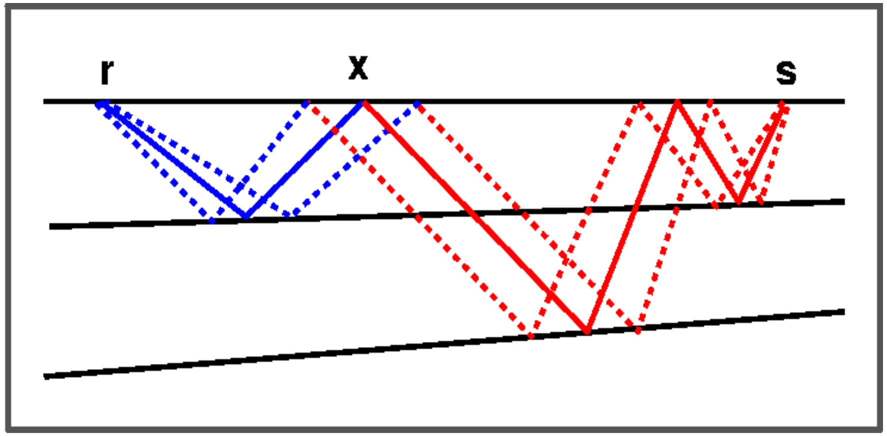
Nordkapp Basin, Barents Sea



Basic principles of SRME

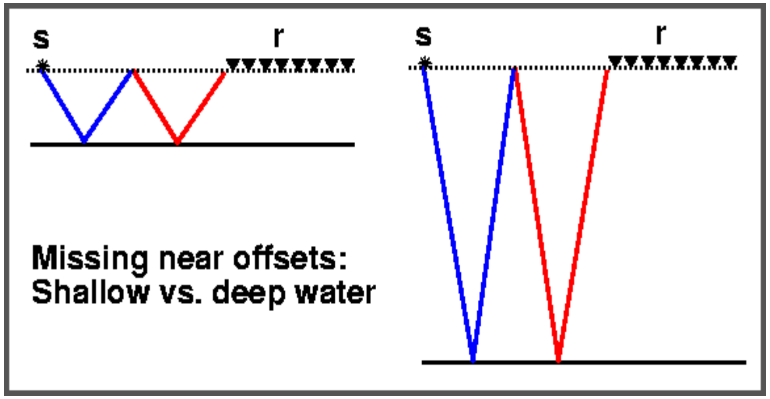


1st order multiple

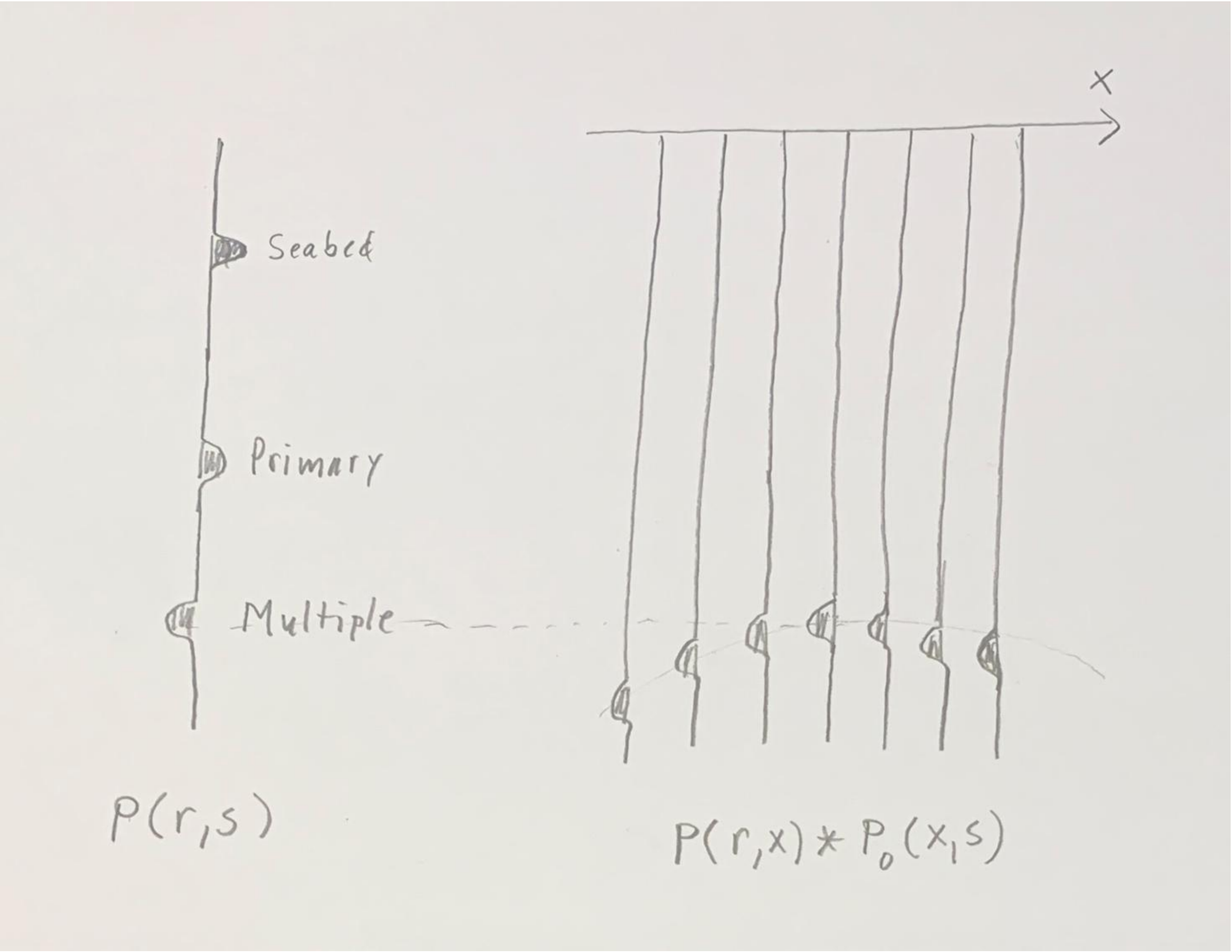


2nd order multiple

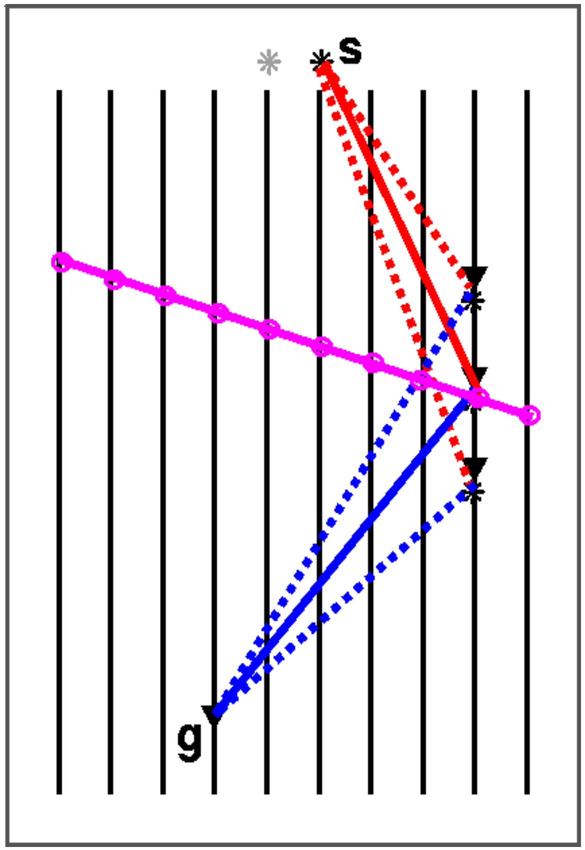
- Multiple prediction by convolution of events
- The specular ray is unknown
- Kirchhoff-migration-like sum
- Adaptive subtraction of predicted multiples



Missing near offsets is a missing-reflection-angle problem


x
Seabed
Primary
Multiple
P(r,s)
P(r,x) * P0(x,s)

Extension to 3D



3D SRME in map view

Sequence of steps:

1. Inline sum along cables (like 2D)
2. X-line sum across cables
3. Adaptive multiple subtraction

- Theoretically 3D SRME is a simple extension of 2D …
- … but 3D seismic acquisition violates theoretical requirements

3D SRME equations

Surface related multiple attenuation (SRME):

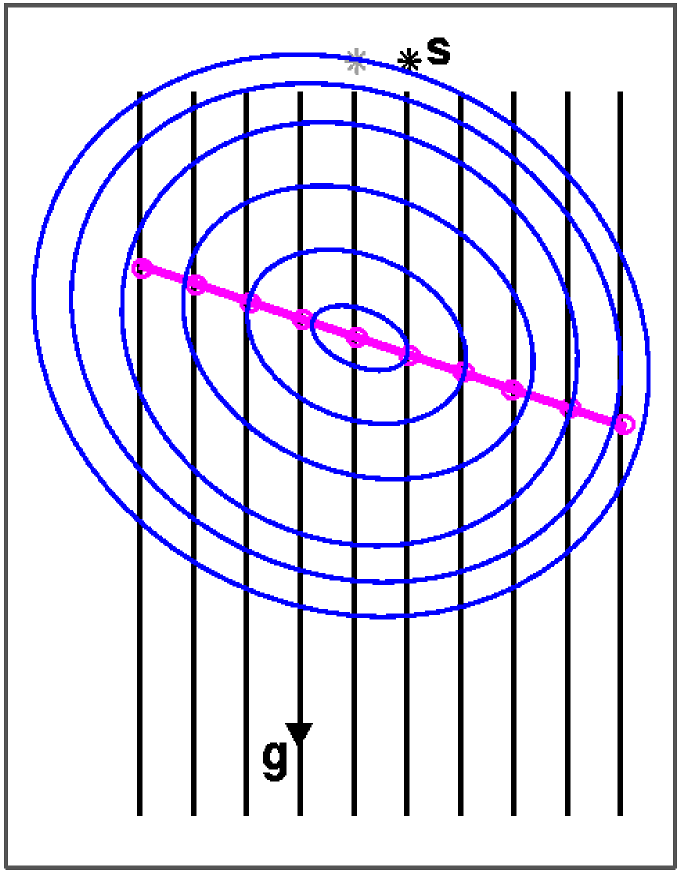
$$P_0(\mathbf{x}_g, \mathbf{x}_s) = P(\mathbf{x}_g, \mathbf{x}_s) - A(\omega) M(\mathbf{x}_g, \mathbf{x}_s)$$

Multiple prediction:

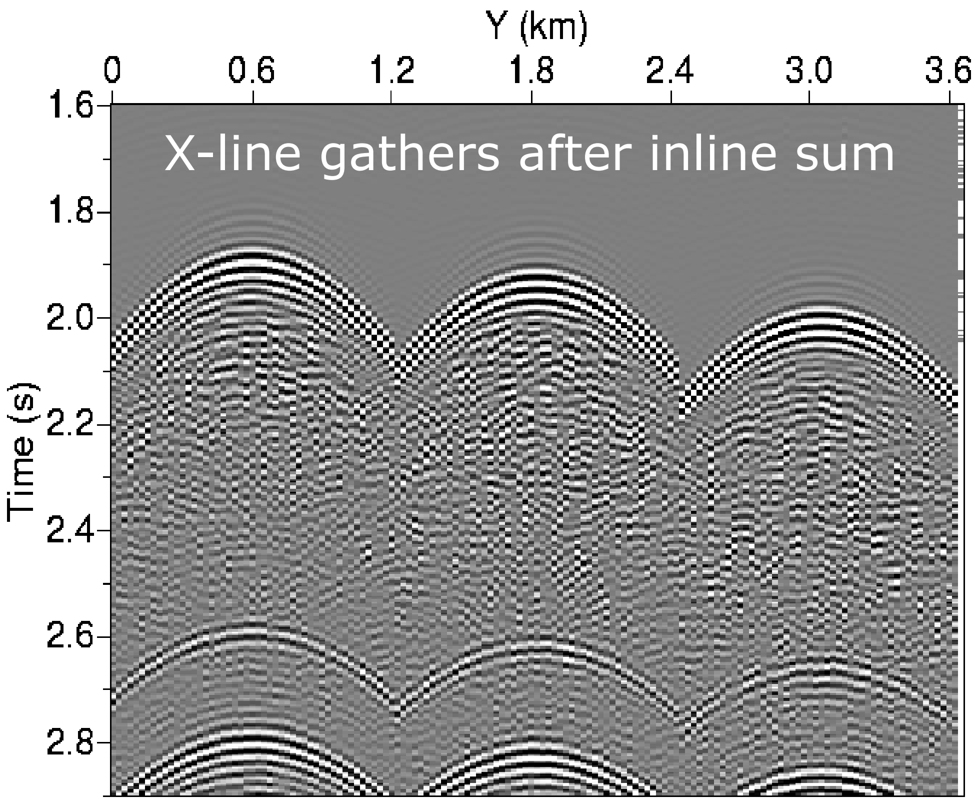
$$\begin{aligned} M(\mathbf{x}_g, \mathbf{x}_s) &= \sum_{y_k} \left[\sum_{x_k} P_0(\mathbf{x}_g, \mathbf{x}_k) P(\mathbf{x}_k, \mathbf{x}_s) \Delta x \right] \Delta y \\ &\simeq \sum_{y_k} \left[\sum_{x_k} P(\mathbf{x}_g, \mathbf{x}_k) P(\mathbf{x}_k, \mathbf{x}_s) \Delta x \right] \Delta y \end{aligned}$$

- The sum over x_k is easy (2D SRME along cables)
- The hard part is the sum over y_k (across cables)
- Solve by sparse inversion

3D vs. 2D SRME



3D SRME in map view



- 2D SRME picks the local inline apex.
- The 2D prediction is always too late (Fermat's principle)

Radon transform

Sum over travel time curves in data $D(t, y)$:

$$M(\tau, q) = \int dy\, D(t = \phi(\tau, q, y), y)$$

Linear, parabolic, hyperbolic transforms:

$$\begin{aligned}
\phi(\tau, q, y) &= \tau + qy \quad \text{(linear; slant stack)} \\
\phi(\tau, q, y) &= \tau + qy^2 \quad \text{(parabolic)} \\
\phi(\tau, q, y) &= \sqrt{\tau^2 + qy^2} \quad \text{(hyperbolic)}
\end{aligned}$$

Model-space parameters: (τ, q)

Parabolic apex-shifted Radon transform

Traveltime function

$$\phi(\tau, q, y^0, y) \quad = \quad \tau + q(y - y^0)^2$$

Model-space parameters: (τ, q, y^0)

Frequency domain:

$$\frac{1}{\sqrt{2\pi}} \int D(\tau + q(y - y^0)^2, y) e^{i\omega\tau} d\tau \quad = \quad D(y; \omega) e^{i\omega q (y - y^0)^2}$$

Frequency-domain apex-shifted parabolic Radon:

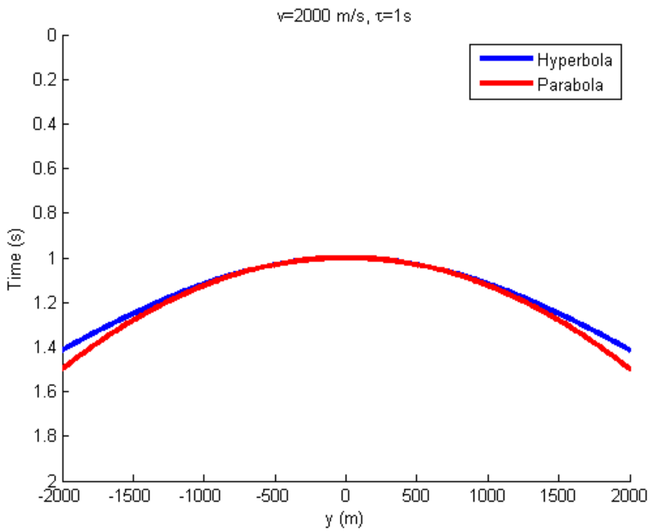
$$M(q, y^0; \omega) \quad = \quad \int D(x; \omega) e^{i\omega q (y - y^0)^2} dy$$

Data vector for sparse inversion:

$$D(y_k;\omega) = \sum_{x_k} P(\mathbf{x}_g, \mathbf{x}_k) P(\mathbf{x}_k, \mathbf{x}_s) \Delta x$$

Parabolic approximation for the crossline multiple:

$$T = \sqrt{\tau^2 + \frac{(y_k - y^0)^2}{\hat{v}^2}} \simeq \tau + q(y^0 - y_k)^2$$



Parabolic Radon transform:

$$M(q_i, y_j^0; \omega) = \sum_{k=1}^{N} D(y_k;\omega) e^{-i\omega q_i (y_k - y_j^0)^2} \Delta y$$

On matrix form

$$\mathbf{m} = \mathbf{L}^H \mathbf{d} \quad \Rightarrow \quad \mathbf{d} = \mathbf{L}\mathbf{m}$$

Formulate it as a sparse-inversion problem for $\mathbf{m}$
(naive direct inverse doesn't work)

Gauss-Cauchy objective function

$$
\begin{aligned}
\Phi &= \frac{1}{2\sigma_n^2}(\mathbf{d}-\mathbf{Lm})^H(\mathbf{d}-\mathbf{Lm}) + C(\mathbf{m}) \\
C &= \sum_{j=1}^{N_m} \ln\left(1 + \frac{|m_j|^2}{2\sigma_m^2}\right)
\end{aligned}
$$

- Gaussian likelihood, Cauchy prior $C(\mathbf{m})$
- Cauchy prior makes the inverse problem non-linear

$$
\begin{aligned}
\frac{\partial \Phi}{\partial \mathbf{m}} &= -\frac{1}{\sigma_n^2}\mathbf{L}^H(\mathbf{d}-\mathbf{Lm}) + \frac{\partial C}{\partial \mathbf{m}} = 0 \\
\frac{\partial C}{\partial m_j} &= \frac{1}{1+\frac{|m_j|^2}{2\sigma_m^2}}\frac{m_j}{\sigma_m^2} = Q_{jj}^{-1}\frac{m_j}{\sigma_m^2} \\
\frac{\partial \Phi}{\partial \mathbf{m}} &= -\frac{1}{\sigma_n^2}\mathbf{L}^H(\mathbf{d}-\mathbf{Lm}) + \frac{1}{\sigma_m^2}\mathbf{Q}^{-1}\mathbf{m} = 0
\end{aligned}
$$

Least-squares solution

$$\mathbf{m} = (\mathbf{L}^H\mathbf{L} + \lambda\mathbf{Q})^{-1}\mathbf{L}^H\mathbf{d}$$

Minimum-norm solution

$$\mathbf{m} = \mathbf{Q}\mathbf{L}^H(\mathbf{L}\mathbf{Q}\mathbf{L}^H + \lambda\mathbf{I})^{-1}\mathbf{d}$$

The matrix $\mathbf{Q}$ accounts for the Cauchy prior

$$Q_{jj} = 1 + \frac{|m_j|^2}{2\sigma_m^2}$$
$$\lambda = \sigma_n^2/\sigma_m^2$$

- Least-squares: dim$(\mathbf{L}^H\mathbf{L}) = N_m \times N_m \sim 500^2$
- Minimum-norm: dim$(\mathbf{L}\mathbf{Q}\mathbf{L}^H) = N_d \times N_d \sim 10^2$

Introduce auxilliary variable $\mathbf{b}$

$$\mathbf{m} = \mathbf{Q}\mathbf{L}^H(\mathbf{L}\mathbf{Q}\mathbf{L}^H + \lambda\mathbf{I})^{-1}\mathbf{d} = \mathbf{Q}\mathbf{L}^H\mathbf{b}$$

Iterative solution (by Cholesky decomposition)

$$\begin{aligned}
\mathbf{b}^{(n-1)} &= (\mathbf{L}\mathbf{Q}^{(n-1)}\mathbf{L}^H + \lambda\mathbf{I})^{-1}\mathbf{d} \\
\mathbf{m}^{(n)} &= \mathbf{Q}^{(n-1)}\mathbf{L}^H\mathbf{b}^{(n-1)} \\
Q_{jj}^{(n)} &= 1 + \frac{|m_j^{(n)}|^2}{2\sigma_m^2}
\end{aligned}$$

Initial value

$$\mathbf{Q}^{(0)} = \mathbf{I}$$

- First iteration is Gauss-Gauss
- 3-5 iterations needed

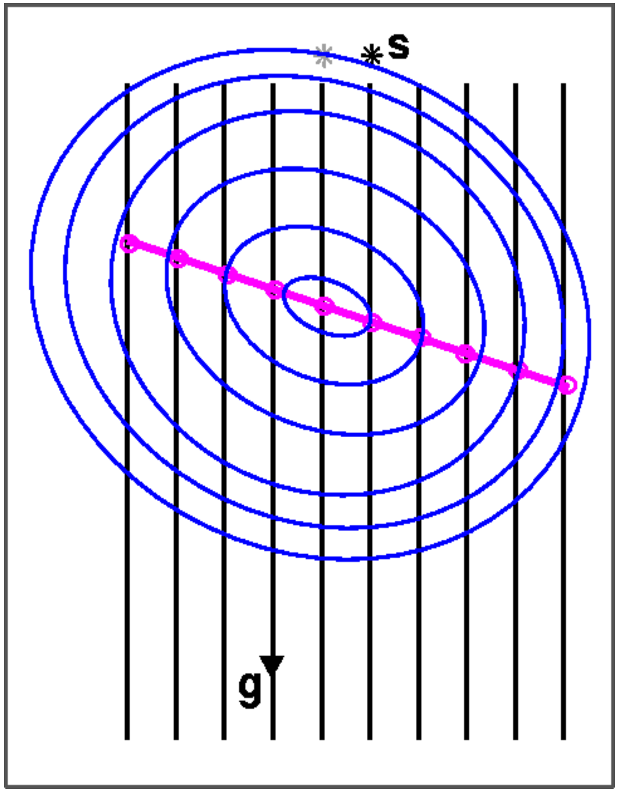
Multiple prediction operator

$$
\begin{aligned}
M(\mathbf{x}_g, \mathbf{x}_s) &= \sum_{j=1}^{N_{y0}} \sum_{i=1}^{N_q} F_c(q_i) m(q_i, y_j^0) \\
F_c(q_i) &= \sqrt{\pi/\omega q_i} e^{-i\pi/4}
\end{aligned}
$$

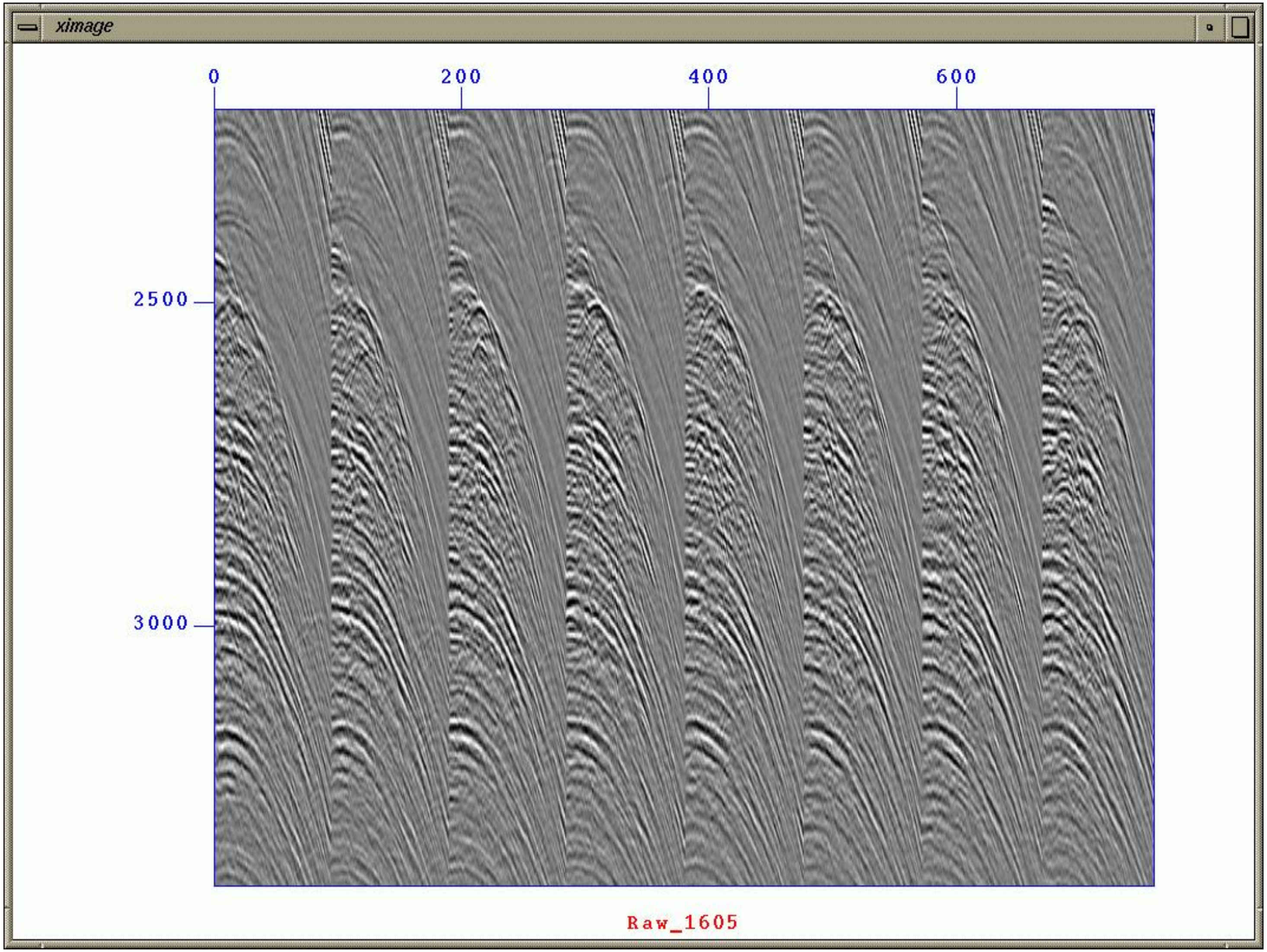
(by Parseval's theorem)

Multiple subtraction:

$$
\begin{aligned}
P_0(\mathbf{x}_g, \mathbf{x}_s) &= P(\mathbf{x}_g, \mathbf{x}_s) - A(\omega) M(\mathbf{x}_g, \mathbf{x}_s) \\
A(\omega) &= \text{inverse source spectrum}
\end{aligned}
$$

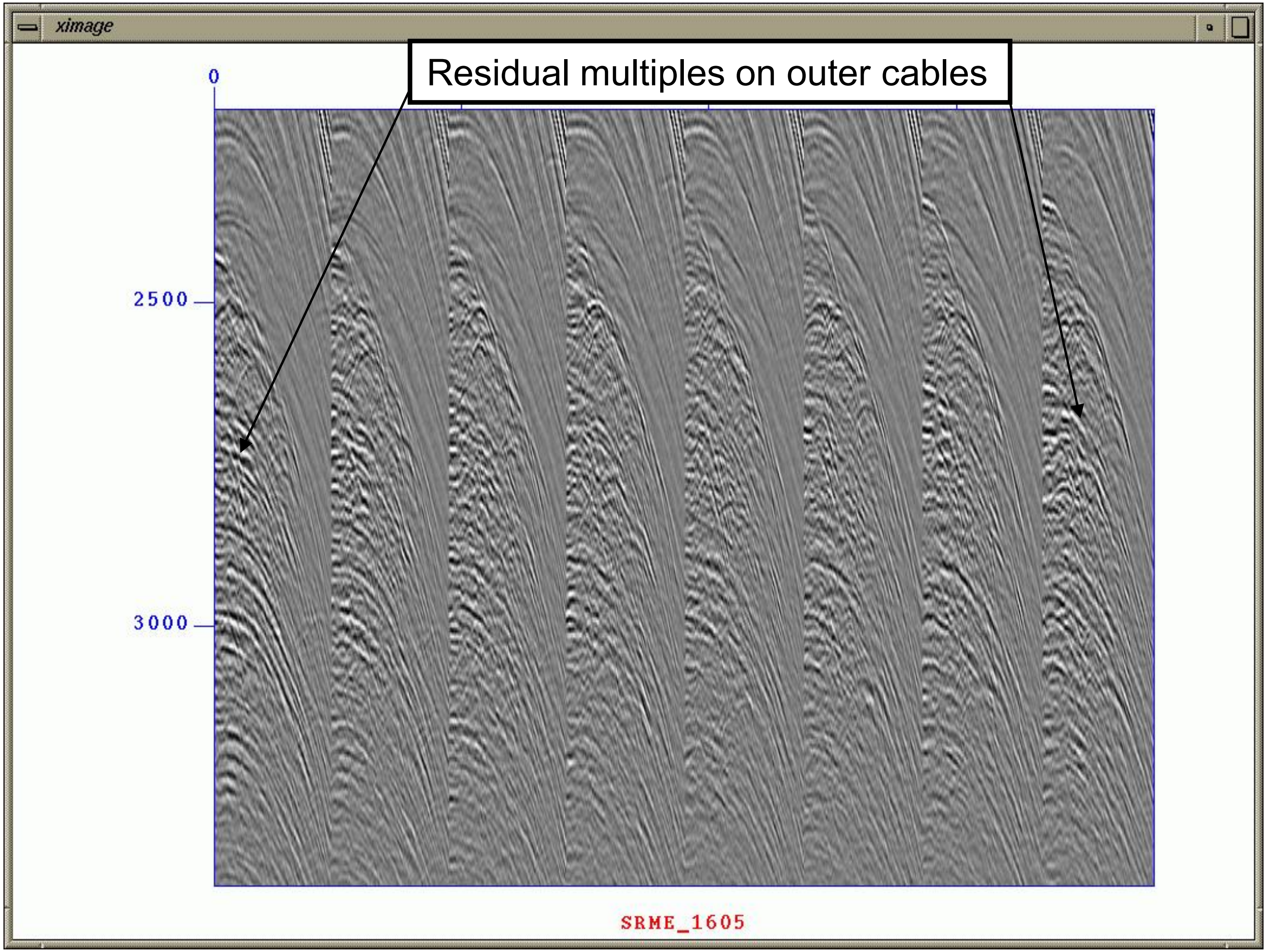


Input data



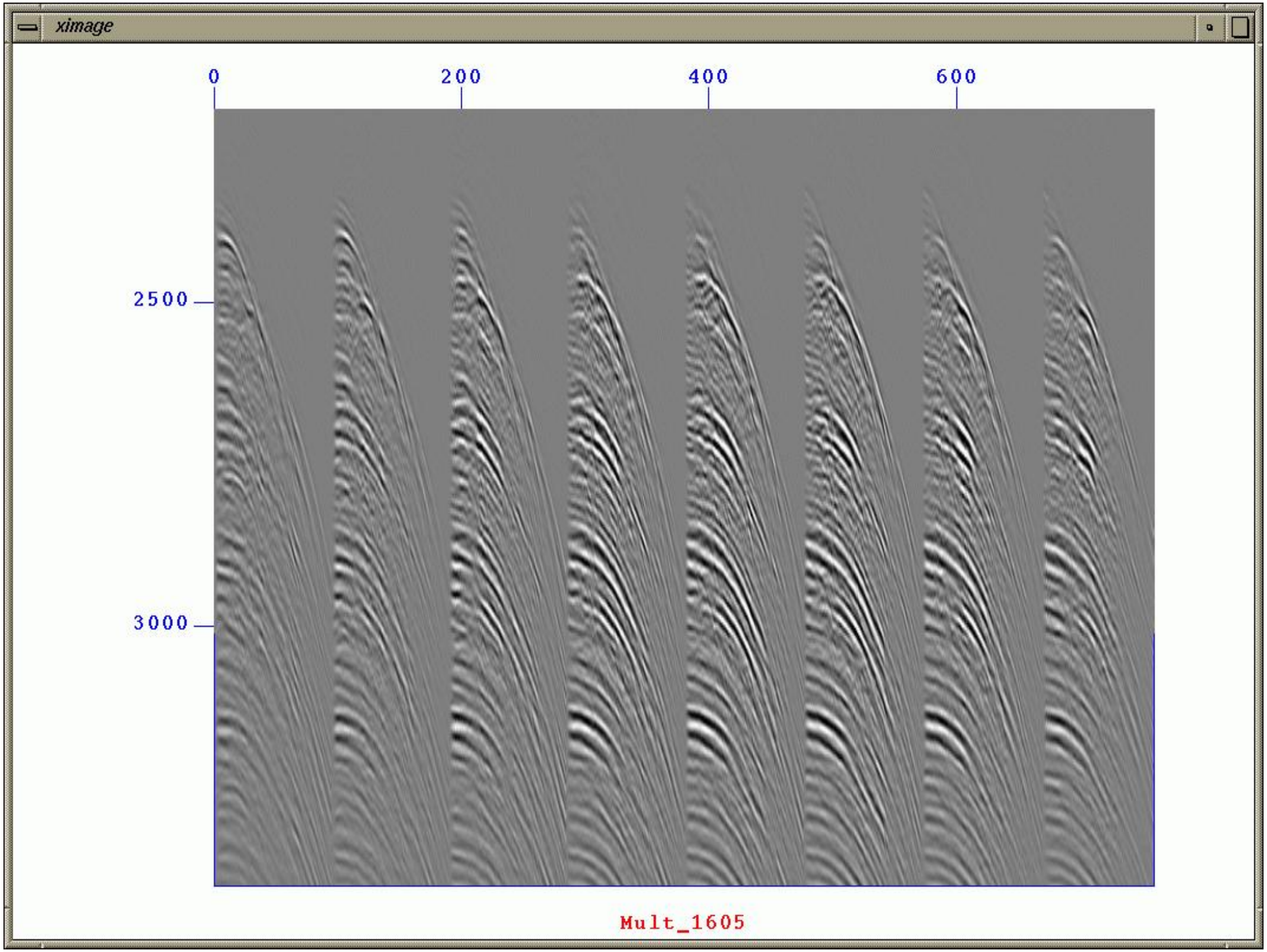
FFID 1605: 8 cables

3D SRME: 400m sail-line separation

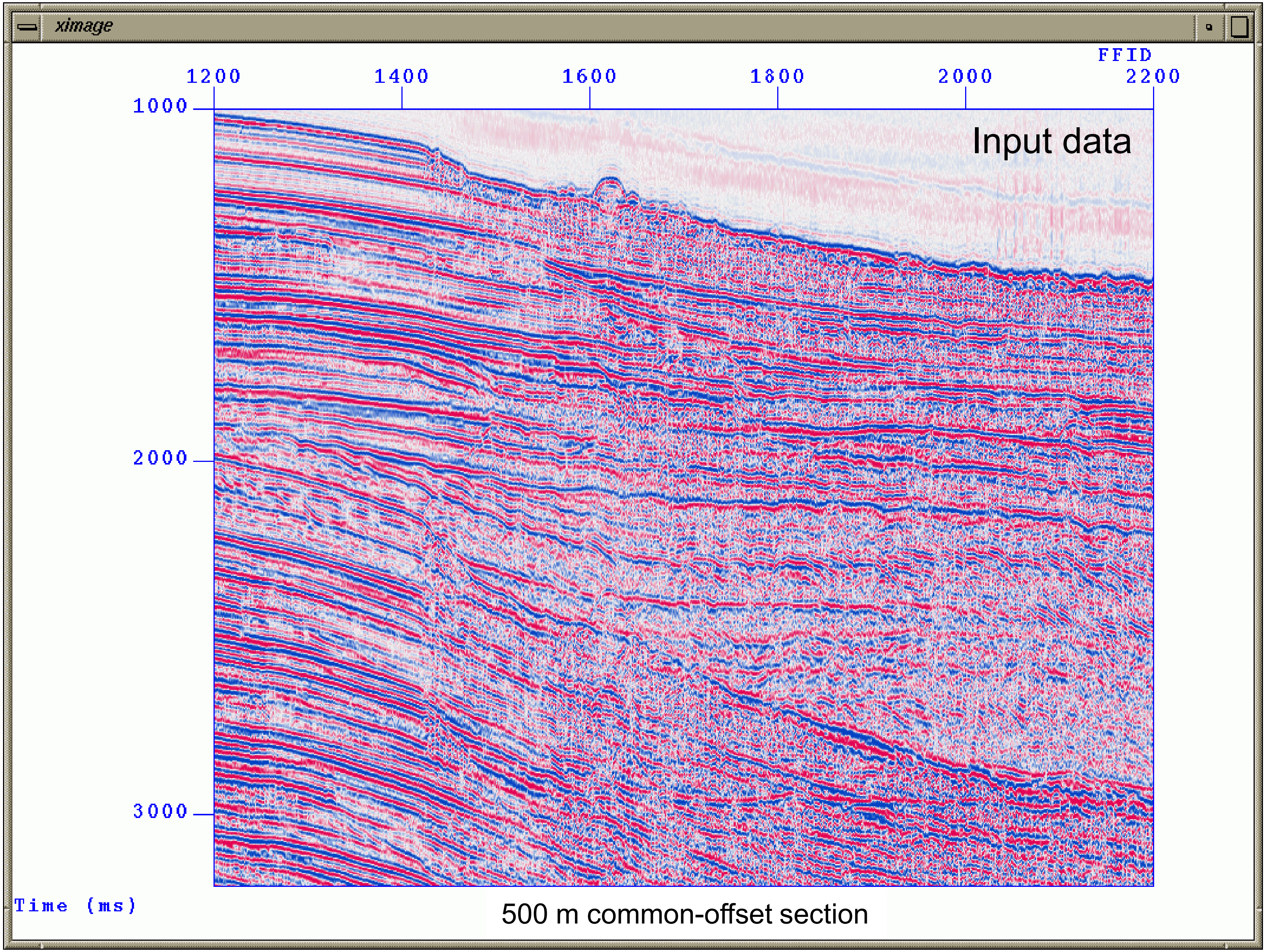


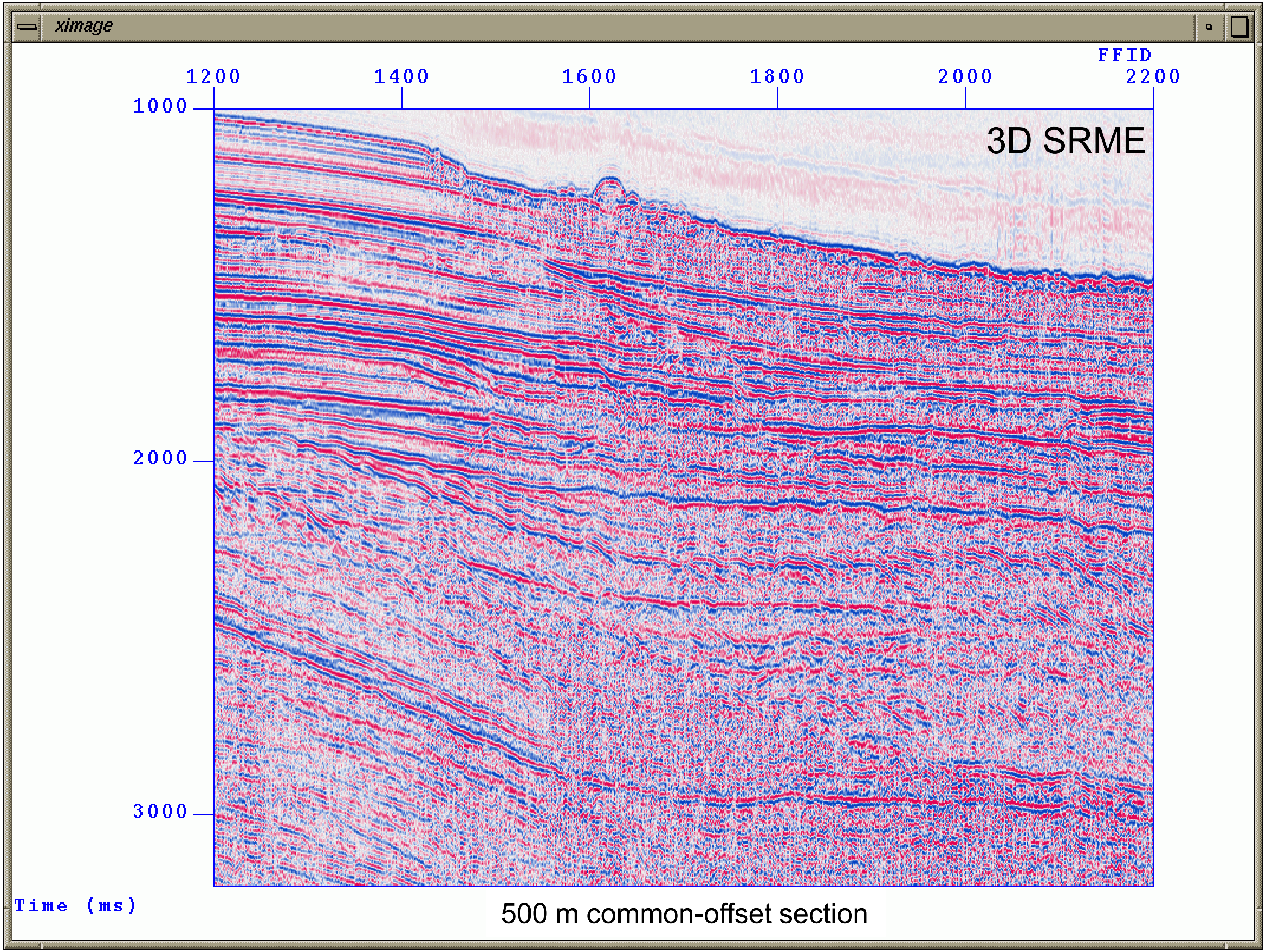
FFID 1605: 8 cables

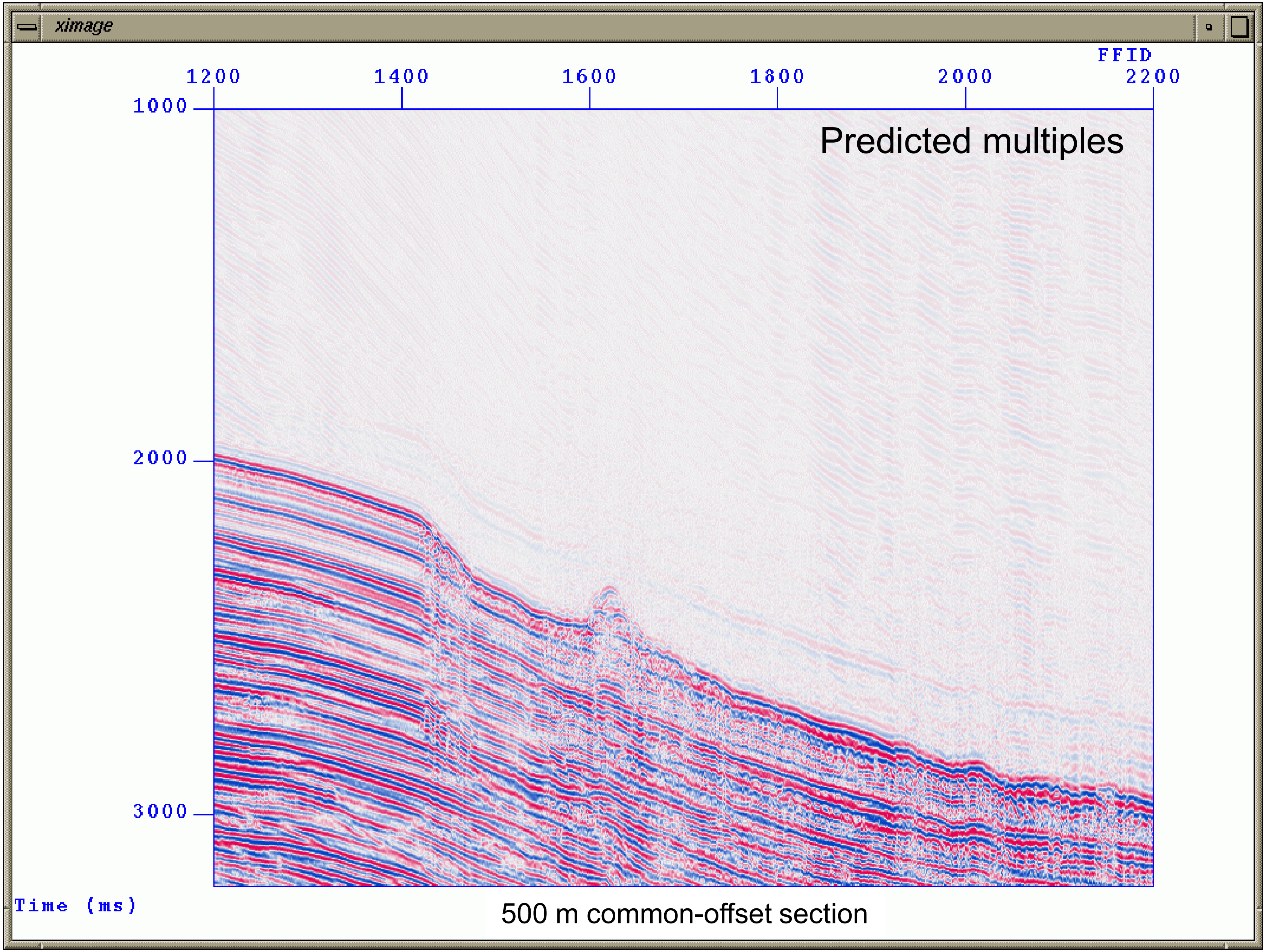
Predicted multiples: 400m sail-line sep.



FFID 1605: 8 cables


ximage
FFID
1200
1400
1600
1800
2000
2200
1000
2000
3000
Input data
Time (ms)
500 m common-offset section


ximage
FFID
1200
1400
1600
1800
2000
2200
1000
2000
3000
3D SRME
Time (ms)
500 m common-offset section


ximage
FFID
1200
1400
1600
1800
2000
2200
1000
2000
3000
Predicted multiples
Time (ms)
500 m common-offset section